Equal Upper Orlicz Index Does Not Force Coarse or Uniform Embeddability

Candidate full counterexample for arXiv:1207.3967

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Status. Candidate full counterexample, likely valid. This packet answers the source question for every prescribed common index $p > 2$, subject to human review.

1 Source question

Kraus studies coarse and uniform embeddings between Orlicz sequence spaces h_M and h_N [1]. On page 11, after treating all other index configurations, the paper asks whether there exist Orlicz functions M, N such that

$$2 < \beta_M = \beta_N < \infty$$

and h_M does not coarsely (uniformly) embed into h_N.

that h_M does not coarsely or uniformly embed into ℓ_p. □

The last remaining case is when $2 < \beta_M = \beta_N < \infty$. In this case, we can of course always have the coarse (uniform) embeddability (since any Banach space strongly uniformly embeds into itself). However, we do not know whether there exist Orlicz functions M, N satisfying $2 < \beta_M = \beta_N < \infty$, such that h_M does not coarsely (uniformly) embed into h_N.

Let us conclude with a brief summary of the results. Let M, N be Orlicz functions.

(1) If $\beta_M < \beta_N$ or $\beta_N \leq \beta_M < 2$ or $\beta_M = \beta_N = \infty$, then h_M strongly uniformly embeds into h_N.
(2) If $\beta_M > 2$ and $\beta_N < \beta_M$, then h_M does not coarsely or uniformly embed into h_N.
(3) If $\beta_N \leq \beta_M = 2$, then the coarse (uniform) embeddability of h_M into h_N is not determined by the values of β_M and β_N.
(4) If $2 < \beta_M = \beta_N < \infty$, then the question of the coarse (uniform) embeddability of h_M into h_N is open.

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References

Figure 1: The open question and summary on page 11 of arXiv:1207.3967.

Transcription. "However, we do not know whether there exist Orlicz functions M, N satisfying $2 < \beta_M = \beta_N < \infty$, such that h_M does not coarsely (uniformly) embed into h_N."

2 Definitions and the metric-cotype input

An Orlicz function is a continuous, nondecreasing convex function $M : [0, \infty) \to [0, \infty)$ with $M(0) = 0$ and $M(t) \to \infty$. Its sequence space h_M is equipped with the Luxemburg norm

$$\|x\|_M = \inf \left\{ \rho > 0 : \sum_i M\left(\frac{|x_i|}{\rho}\right) \le 1 \right\}.$$

For the present normalization, the upper Matuszewska–Orlicz index is

$$\beta_M = \inf \left\{ q : \inf_{\lambda, t \in (0,1]} \frac{M(\lambda t)}{M(\lambda) t^q} > 0 \right\}.$$

We use the following specialization of the optimal-scale metric-cotype theorem of Mendel and Naor [2, Theorem 4.1]. It is important here that ℓ_p is K-convex and has cotype p.

Lemma 1 (Metric cotype of ℓ_p)**.** *For every $p > 2$ there are constants $\Gamma, C < \infty$ such that for each $n \ge 1$ one can choose an integer $m = m_n$, divisible by four, satisfying*

$$C^{-1} n^{1/p} \le m \le C n^{1/p},$$

and every map $f : \mathbb{Z}_m^n \to \ell_p$ satisfies

$$\sum_{j=1}^{n} \mathbb{E}_x \left\| f(x + \tfrac{m}{2} e_j) - f(x) \right\|_p^p \le \Gamma^p m^p \mathbb{E}_{x,\varepsilon} \left\| f(x + \varepsilon) - f(x) \right\|_p^p, \tag{1}$$

where x is uniform on $\mathbb{Z}_m^n$ and ε is uniform on $\{-1, 0, 1\}^n$.

Proof. Apply Mendel–Naor's theorem with both of its exponents equal to p and $X = \ell_p$. It gives a constant threshold $An^{1/p}$ and a uniform metric-cotype constant for every multiple of four above that threshold. Choose the least such multiple above $\max\{4, An^{1/p}\}$. This choice is also at most a constant times $n^{1/p}$. □

3 The counterexample

Fix $p > 2$ and define

$$M(0) = 0, \qquad M(t) = \frac{t^p}{1 - \log t} \quad (0 < t \le 1), \qquad M(t) = 1 + (p+1)(t-1) \quad (t \ge 1). \tag{2}$$

Let $N(t) = t^p$.

Theorem 1. *For every $p > 2$, the functions M, N in (2) satisfy*

$$\alpha_M = \beta_M = \beta_N = p,$$

but h_M admits neither a coarse embedding nor a uniform embedding into $h_N = \ell_p$. Consequently the page-11 question of Kraus has an affirmative answer for every possible common value $p > 2$.

4 Proof intuition

The logarithmic denominator leaves both Matuszewska–Orlicz indices equal to p, but makes the fundamental function of the canonical basis slightly smaller than that of ℓ_p:

$$\Phi_M(n) = \|e_1 + \cdots + e_n\|_M = o(n^{1/p}).$$

To turn this endpoint defect into a nonlinear obstruction, place n circles inside the first $2n$ coordinates of h_M. Moving one coordinate halfway around its circle has fixed size, while moving every coordinate by at most one torus step has size at most $\Phi_M(2n)/m_n = o(1)$. The metric-cotype inequality says that these two scales cannot be separated in this way after a map into ℓ_p. Scaling the circles so short steps have size one defeats a coarse embedding; leaving them unscaled defeats a uniform embedding.

5 Proof of Theorem 1

Step 1: the Orlicz function and its indices

Put $L(t) = 1 - \log t$. On $(0, 1]$ a direct differentiation gives

$$M'(t) = t^{p-1}\left(\frac{p}{L(t)} + \frac{1}{L(t)^2}\right) > 0, \qquad M''(t) = \frac{t^{p-2}}{L(t)^3}\left[p(p-1)L(t)^2 + (2p-1)L(t) + 2\right] > 0.$$

Also $M'(1^-) = p + 1$, so the tangent linear continuation in (2) is convex. Thus M is an Orlicz function.

For $\lambda, t \in (0, 1]$,

$$\frac{M(\lambda t)}{M(\lambda)t^q} = t^{p-q}\frac{L(\lambda)}{L(\lambda t)}. \tag{3}$$

If $q \leq p$, the right side is at most $t^{p-q} \leq 1$, which yields $\alpha_M \geq p$. If $q > p$, then $L(\lambda) \geq 1$ and

$$t^{p-q}\frac{L(\lambda)}{L(\lambda t)} = \frac{t^{p-q}}{1 + (-\log t)/L(\lambda)} \geq \frac{t^{p-q}}{1 - \log t}.$$

The last function is positive and continuous on $(0, 1]$ and tends to ∞ at zero, so it has a positive infimum. Hence $\beta_M \leq p$. Since always $\alpha_M \leq \beta_M$, we have $\alpha_M = \beta_M = p$. Clearly $\beta_N = p$ and $h_N = \ell_p$.

Let

$$\Phi(k) = \|e_1 + \cdots + e_k\|_M = \frac{1}{M^{-1}(1/k)}.$$

Writing $s_k = M^{-1}(1/k)$, we have $k^{-1} = s_k^p/L(s_k)$ and therefore

$$\frac{\Phi(k)}{k^{1/p}} = \frac{1}{L(s_k)^{1/p}} \longrightarrow 0. \tag{4}$$

Step 2: a family of discrete tori in h_M

Let $m = m_n$ be supplied by Lemma 1. For $a > 0$ define $G_{n,a} : \mathbb{Z}_m^n \to h_M$ by

$$G_{n,a}(x) = a\sum_{j=1}^{n}\left[\cos\left(\frac{2\pi x_j}{m}\right)e_{2j-1} + \sin\left(\frac{2\pi x_j}{m}\right)e_{2j}\right].$$

Shifting x_j by $m/2$ changes the corresponding two-coordinate vector to its negative. Since one of the absolute values of its sine and cosine is at least $1/\sqrt{2}$, lattice monotonicity of the Luxemburg norm gives

$$\left\|G_{n,a}(x+\tfrac{m}{2}e_j)-G_{n,a}(x)\right\|_M \geq \sqrt{2}\,a. \tag{5}$$

The sine and cosine functions are one-Lipschitz. Hence, for every $\varepsilon \in \{-1,0,1\}^n$, each of the first $2n$ coordinates of $G_{n,a}(x+\varepsilon)-G_{n,a}(x)$ has absolute value at most $2\pi a/m$. Again by lattice monotonicity and symmetry,

$$\|G_{n,a}(x+\varepsilon)-G_{n,a}(x)\|_M \leq \frac{2\pi a}{m}\Phi(2n). \tag{6}$$

Step 3: no coarse embedding

Suppose $F : h_M \to \ell_p$ were a coarse embedding. Let $\rho : [0,\infty) \to [0,\infty)$ be a nondecreasing lower control with $\rho(t) \to \infty$, and let

$$W = \sup\{\|F(u)-F(v)\|_p : \|u-v\|_M \leq 1\} < \infty.$$

Set

$$a_n = \frac{m_n}{2\pi\Phi(2n)}.$$

By (4) and $m_n \asymp n^{1/p}$, we have $a_n \to \infty$. Equations (5)–(6) and Lemma 1, applied to $F \circ G_{n,a_n}$, give

$$n\rho(\sqrt{2}a_n)^p \leq \Gamma^p m_n^p W^p.$$

Thus

$$\rho(\sqrt{2}a_n) \leq \Gamma \frac{m_n}{n^{1/p}} W \leq C\Gamma W,$$

contradicting $a_n \to \infty$ and $\rho(t) \to \infty$.

Step 4: no uniform embedding

Suppose instead that $F : h_M \to \ell_p$ were a uniform embedding. Define

$$\omega_F(t) = \sup\{\|F(u)-F(v)\|_p : \|u-v\|_M \leq t\}.$$

Uniform continuity gives $\omega_F(t) \to 0$ as $t \downarrow 0$ (and makes $\omega_F(t)$ finite). Uniform continuity of F^{-1} on $F(h_M)$ gives an $\eta > 0$ such that

$$\|u-v\|_M \geq \sqrt{2} \quad \Longrightarrow \quad \|F(u)-F(v)\|_p \geq \eta. \tag{7}$$

Take $a = 1$ in $G_{n,a}$. By (4),

$$\delta_n = \frac{2\pi\Phi(2n)}{m_n} \longrightarrow 0.$$

Applying Lemma 1 to $F \circ G_{n,1}$ and using (5), (6), and (7), we obtain

$$n\eta^p \leq \Gamma^p m_n^p \omega_F(\delta_n)^p.$$

Consequently

$$\eta \leq \Gamma \frac{m_n}{n^{1/p}} \omega_F(\delta_n) \longrightarrow 0,$$

a contradiction. This completes the proof. $\square$

6 Verification and limitations

- The Orlicz convexity and index computations are explicit. The same logarithmic function was used by Kraus at the Hilbert endpoint $p = 2$; here it is used for every $p > 2$.
- The only external mathematical input is the optimal $m_n \asymp n^{1/p}$ metric-cotype inequality for ℓ_p, obtained from [2, Theorem 4.1]. Merely knowing the cotype exponent would not suffice.
- No numerical experiment is used. The proof is scale- and constant-robust.
- A bounded novelty check on 9 August 2026 searched the exact source sentence, the paper title and arXiv id, and combinations of "Orlicz sequence space", "coarse embedding", "uniform embedding", equal upper index, and Matuszewska–Orlicz. It located the source and general later embedding work, but no later paper claiming this endpoint answer. This is not a proof of novelty.
- Human review should focus on the specialization of Mendel–Naor's Theorem 4.1 and on the two modulus arguments after (5)–(6).

References

[1] M. Kraus, *Coarse and uniform embeddings between Orlicz sequence spaces*, Mediterr. J. Math. 11 (2014), 653–666; arXiv:1207.3967.

[2] M. Mendel and A. Naor, *Metric cotype*, Ann. of Math. (2) 168 (2008), 247–298; arXiv:math/0506201.